

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

Keylock

Version 2.61 Feb. 2008

Notes

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Section 1. Introduction

This manual describes the method of use and related items, as well as device-specific precautions, when the Keylock devices are used with EPSON OPOS ADK.

Before the KeyLock device can be used, the EPSON OPOS ADK should be installed and the devices to be used should be set using the SetupPOS utility. For setting methods, see the Section 2 of this manual.

This manual applies to the following devices.

- EPSON DM-KX060

The operation of Keylock is not supported in Windows Vista environment.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for the KeyLock devices.

2.1 Device Information

The DeviceDescription and DeviceName for each model are as follows.

Model Name	DeviceDescription	DeviceName
EPSON DM-KX060	EPSON DM-KX060 Keylock	DM-KX060

2.2 Hardware Settings

It is a prerequisite for the operation of this OCX that the following codes are correctly set. Before use, use the utilities provided with each device to check that the following settings have been made correctly.

2.2.1 EPSON DM-KX060

Key Position	Scan Code
PRG	64h
REG	65h
MA1	66h
X	67h
Z	68h
MA2	69h
S	6Ah
OFF	6Bh

2.3 Setting of Devices

After confirming the model number of the hardware, select the correct device class using the SetupPOS utility. Regarding how to use the SetupPOS utility, refer to the "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".

2.4 EPSONHKO (Hook Control Utility)

The Keylock device data are all sent to the PC as keyboard data. The hook driver utility is a DLL that separates the data to every device, the MSR, the Keylock, and the Keyboard. When the Keylock device's OCX is active (from Open to Close), the hook driver is automatically called up for operation. The incorporation of the EPSONHKO (keyboard hook control utility) also allows the hook control utility to be loaded and operated when the hook driver is required at any other time. The EPSONHKO can be incorporated by the communications setting of the SetupPOS utility. For the EPSONHKO, refer to "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".

*The hook driver Version 1.xx and Version 2.xx cannot operate simultaneously.

Section 3. Function Details

This section describes the functions of the KeyLock device in details. Supplementary explanation of the parts not described in detail in "UPOS" is also given here.

3.1 CheckHealth Method

3.1.1 Internal Test

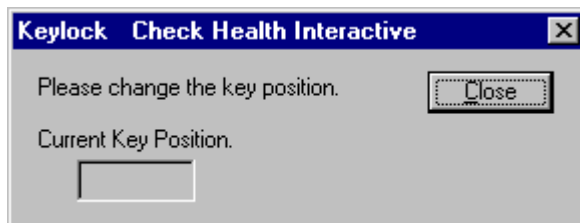
Not supported

3.1.2 External Test

Not supported

3.1.3 Interactive Test

The following dialog box is displayed and the current key position is indicated. Information on the key position changes each time the key position is changed. When the key position is unknown, nothing is displayed.



When executed by OPOS_CH_INTERACTIVE, the character strings of the CheckHealthText property are as follows.

- “Interactive Hcheck: Canceled” : When the [CLOSE] button is pressed without anything being displayed for the current key position.
- “Interactive Hcheck: Complete” : When CheckHealth has been executed according to the messages.

3.2 Device Specific Settings

Whether or not to issue a key event for the OFF position can be set in the [Device Specific Settings] dialog box via SetupPOS utility.



By selecting or clearing the [Return the key OFF position.] check box, the user may specify whether or not to issue a key event when the key is turned to the OFF position.

Select : Issue a key event.

Clear : Do not issue a key event.

3.3 Property Set Values and Default Values

3.3.1 Capability Set Value

Capability name	All devices
CapPowerReporting	0 (OPOS_PR_NONE)
CapKeylockType	1 (KT_STANDARD)

3.3.2 Property Default set Values and Setting Ranges

3.3.2.1 EPSON DM-KX060

KeyPosition	1, 2, 3, 4, 5, 6, 7, 8
PositionCount	8
ElectronicKeyValue	""

The KeyPosition is initialized according to the key position of the device when the DeviceEnabled property becomes TRUE. If the key position of the device is not changed at least once after the hook driver control is started, the hook driver will not be able to grasp the key position, and it will be initialized to "–1".

DM-KX060 is equipped with a feature that enables the user to obtain the current key position. When the current key position is successfully obtained, then the KeyPosition shall be initialized according to the newly obtained data. However, if the current key position cannot be successfully obtained and should an error result, the same initialization process shall follow as other devices.

3.4 Device Statistics

The DeviceStatistics function is added in response to the compliance of the "UPOS 1.8".

Please refer to the "EPSON OPOS ADK MANUAL APPLICATION GUIDE Device Statistics" for the details of the Device Statistics.

Section 4. Expanded Functions

This section describes the expanded functions of the KeyLock device.

4.1 DirectIO Function

The usage of the DirectIO method and DirectIOEvent event is described as follows.

4.1.1 DirectIO Method

No use

4.1.2 DirectIOEvent Events

No use

Section 5. Device Specific Programming

This section describes the functions specific to the KeyLock device.

5.1 Key Position

5.1.1 DM-KX060

Eight key positions are available for EPSON DM-KX060.

Details are as follows:

Position	OPOS Definition	Value	KX060 Settings Code	Scan Code
PRG	LOCK_KP_NORM	2	08h	64h
OFF	None	8	40h	6Bh
REG	LOCK_KP_LOCK	1	10h	65h
MA1	None	4	18h	66h
X	None	5	20h	67h
Z	None	6	28h	68h
MA2	None	7	30h	69h
S	LOCK_KP_SUPR	3	38h	6Ah

The scan codes can be changed using the utility provided with EPSON DM-KX060. However, note that if the values are changed to others than those mentioned above (defaults), the Keylock control would not work correctly. When setting scan codes, be sure to do so while referring to the product manual of the device.

Section 6. Error Information

This section describes the error codes that may result from execution of KeyLock methods. The common properties and methods are described in "EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE GENERAL DEVELOPMENT". Refer to this guide for more information.

6.1 ResultCode List

The ResultCode and ResultCodeExtended when methods are executed are as follows.

Method Name	ResultCode	ResultCodeExtended	Meaning
WaitForKeylockChange	OPOS_SUCCESS	0	Refer to UPOS Specifications.
	OPOS_E_CLOSED	0	Refer to UPOS Specifications.
	OPOS_E_DISABLED	0	Refer to UPOS Specifications.
	OPOS_E_ILLEGAL	OPOS_EX_BADPARAM + 1	First parameter is illegal.
		OPOS_EX_BADPARAM + 2	Second parameter is illegal.
	OPOS_E_TIMEOUT	0	Refer to UPOS Specifications.

6.2 Remedial Actions for Principal Errors

ResultCodeExtended	Remedy
OPOS_EX_BADINF	Normally, this should not occur. It occurs if the INF file or the Registry information of the device to be used is incorrect. Using the INF file provided with EPSON OPOS ADK, perform a new setup using the SetupPOS utility.
OPOS_ELOCK_KBHOOKOPEN OPOS_ELOCK_KBHOOKSTART OPOS_ELOCK_KBHOOKSTOP	Error related to the hook driver. Normally, this should not occur. Should it occur, then use the SetupPOS utility to confirm that the device is a device for which the hook driver can be used.
OPOS_EX_BADPARAM + 1, 2	The parameter is not correct for the method to be used. Confirm the parameter range for the method to be used.

Section 7. Warnings

- Be sure to change the Keylock position. Performing switching operation at high speed may lead to abnormal processing.
- When used simultaneously with MSR, do not change the key position during the interval from when the card reading by the MSR device starts and until it is completed (the LED on MSR lights up, etc.). If the key position is changed while data processing is being performed internally in the MSR, interference between the key position data and MSR data will occur, and the MSR card contents may not be read correctly, or the phenomenon where the KeyPosition property temporarily differs from the actual key position may occur.
- If the EPSONHKO (keyboard hook control utility) is not going to be used, the current key position cannot be determined correctly until the key position has been changed once when the first SO is open status after the OS is started up. In this condition, the KeyPosition property is returned as -1 (key position unknown). To avoid this, it is necessary with some process to prompt the user to change the key position using the application in some kind of way.
When the EPSONHKO is used, a dialog box will request the setting to be changed to LOCK_KP_LOCK at the time the EPSONHKO is started up, and thus the key position can be adjusted to the correct position.
* DM-KX060 comes with a feature that enables it to obtain the correct current key position and, therefore, the above adjustment is not necessary.
- When the EPSONHKO (keyboard hook control utility) is not used, the following phenomena will occur. For the EPSONHKO, refer to "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".
When the process that initiated the hook driver is terminated, Keylock may become inoperable.
For example,
1) Opens Keylock at process 1.
2) Opens Keylock at process 2 and makes the Keylock ready to use by executing DeviceEnabled.

3) Closes Keylock at process 1.

4) Terminates process 1.

5) As the result, Keylock cannot be used at process 2.

To avoid this problem, be sure not to terminate the first opened OCX when other Keylock is opened. Closing the Keylock device is acceptable.

- These Keylock devices do not support power reporting.
- The hook driver Version 1.xx and Version 2.xx cannot operate simultaneously.